

Autosomal dominant polycystic kidney disease

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Autosomal dominant polycystic kidney disease (ADPKD) is the most common genetic cause of chronic kidney disease, resulting in substantial morbidity and mortality globally. Advances in molecular genetics and deep-phenotype imaging techniques have refined existing diagnostic and prognostic tools. The strong evidence-base for tolvaptan as a disease-modifying treatment supports its early use in groups at high risk of kidney failure. Screening and management of potentially serious complications including cyst infection, intracranial aneurysms, and polycystic liver disease are important components of a comprehensive care plan. This Review focuses on current approaches to diagnosis, risk assessment, treatment, and specific aspects of clinical management in ADPKD. An updated understanding of the genetic basis of disease, pathobiology with respect to potential therapeutic targets, and promising therapies now in clinical trials are summarised. We propose a holistic patient-centred care pathway that emphasises shared decision-making with a multidisciplinary clinical team to address the individual needs of patients throughout their lifelong journey.

Introduction

Autosomal dominant polycystic kidney disease (ADPKD) is the most common monogenic cause of chronic kidney disease worldwide, imposing a heavy economic burden on health-care systems.^{1,2} The global prevalence of ADPKD has been estimated to be approximately 1 in 1000 based on *PKD1* and *PKD2* protein-truncating mutations by population whole exome or genome sequencing and approximately 4 in 10000 based on surveys of national clinical registries within the EU and USA.^{3–5} This difference indicates that ADPKD is underdiagnosed. The increase in number and size of cysts with age result in a quasi-exponential expansion of total kidney volume during adult life,⁶ eventually leading to stage 5 chronic kidney disease in approximately half of all patients by age 60 years.⁷ Although only a handful of cysts might be detectable during the teenage years, bilateral kidney enlargement together with innumerable cysts of different sizes can be detected by ultrasound in most patients by their third to fourth decade of life. Typically, the diagnosis of ADPKD is established during asymptomatic screening in the presence of a positive family history, investigation of abdominal or flank pain, haematuria, kidney stones, or upon incidental findings from abdominal imaging.⁸ Hypertension is highly prevalent in most patients with stage 3 chronic kidney disease or beyond and might occur in young adults.^{7–10} Disease progression is commonly associated with haematuria, urinary tract infections, kidney stones, and kidney failure. Non-kidney manifestations such as liver and pancreatic cysts, seminal megavesicles, bronchiectasis, intracranial aneurysms, abdominal hernias, diverticulosis, and cardiovascular abnormalities occur variably.^{11–19} In this Review, we present our approach to management of ADPKD with respect to diagnosis, risk assessment, and treatment, and highlight key concepts from the Kidney Disease: Improving Global Outcomes (KDIGO) 2025 Clinical Practice Guideline for ADPKD.¹¹ We also review pathobiology relevant to therapeutics and discuss promising novel therapeutics currently in clinical trials.

Genetics

Most patients with ADPKD display a dominant inheritance in their family tree; however, approximately 15% will have a negative family history due to de novo disease, and another 10% will have an indeterminate family history due to missing parental records or unrecognised mild disease in an affected parent.²⁰ Mutations of two major genes (*PKD1* and *PKD2*) and several minor genes (*GANAB*, *ALG5*, *ALG8*, *ALG9*, *IFT140*, and *DNAJB11*) account for the majority of genetically resolved adult patients with ADPKD, with *PKD1* mutations being the most common.^{11,21–32} Polycystin-1 (PC1) and polycystin-2 (PC2), localised to the primary cilia and other non-cilia cellular compartments, are proteins encoded by *PKD1* (on chromosome 16) and *PKD2* (on chromosome 4).^{33,34} More than 2500 different *PKD1* and *PKD2* mutations have been reported. This high level of genetic heterogeneity requires comprehensive screening of both genes, which has been facilitated by advances in next-generation sequencing (NGS).^{33,34} In genetically enriched cohorts, approximately 75% of cases are caused by *PKD1* mutations and approximately 15% by *PKD2*

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For more on *PKD1* and *PKD2* mutations see <https://pkdb.mayo.edu/variants>

Search strategy and selection criteria

We searched PubMed for English language articles published between Sept 16, 2018, and April 24, 2025 with the following terms: “adpkd” OR “autosomal dominant polycystic kidney” OR “polycystic kidney disease”, and cross-referenced them with the following terms: “definition”, “genetics”, “diagnosis”, “prognosis”, “management”, “therapy”, and “clinical trials”. All types of studies, including randomised control trials, cohort studies, case-control studies, cross-sectional studies, case reports, systematic reviews, and meta-analyses were considered, but priority was given to studies with high-quality evidence, especially randomised controlled trials, systematic reviews, and large observational studies. We also included older publications based on experience and more recent publications based on reviewer feedback.

mutations.^{22,23} On average, patients with *PKD1* protein-truncating mutations develop a more aggressive phenotype, with larger total kidney volume and earlier progression to stage 5 chronic kidney disease than those with *PKD1* non-truncating or *PKD2* mutations.^{7,22–24} For example, a large cohort study of 220 families from Toronto, ON, Canada reported mean ages of onset of stage 5 chronic kidney disease of 53 years (95% CI 51–54) for *PKD1* protein truncating, 59 years (95% CI 55–62) for *PKD1* in-frame indel, 71 years (95% CI 68–74) for *PKD1* missense, and 80 years (95% CI 77–83) for *PKD2* mutations.²⁴ However, substantial intrafamilial kidney disease variability is well documented, suggesting a strong modifier effect from other genetic and environmental factors.^{35,36} A deleterious genetic modifier effect is supported by the finding of more severe disease in few patients (<50) with digenic mutations (ie, two mutations in each copy of *PKD1* or in one copy of *PKD1* and *PKD2*) compared with their affected relatives carrying a single mutation, including some very early onset cases presenting at birth or in early childhood.^{20,25,37–41} In contrast, a protective modifier effect has been

suggested by the finding of very mild to mild disease among affected relatives in 18% of families with *PKD1* protein-truncating mutations.⁴² Thus, *PKD1* and *PKD2* mutation class alone cannot be used for prognostication with high certainty at the level of the individual patient.³⁴

Up to 15% of patients with ADPKD have no *PKD1* and *PKD2* mutations detected²⁴ and many display an atypical kidney imaging pattern associated with good preservation of kidney function and a low risk for advanced kidney failure.⁴³ A substantial proportion of these patients might have other genetic causes that can be clarified by genetic testing.^{33,34} Mutations in genes known to cause autosomal dominant polycystic liver disease (ADPLD) including *ALG5*, *ALG8*, *ALG9*, *SEC61B*, *SEC63*, *GANAB*, and *PRKCSH* phenocopy mild ADPKD with few to multiple kidney cysts but mild to severe polycystic liver disease.^{11,26–30} Mutations of the ADPLD genes and other minor ADPKD genes could potentially account for some cases of polycystic kidney disease with no mutations detected when only *PKD1* and *PKD2* are screened.^{8,30–32} Additionally, somatic mosaicism has been documented in some patients with a negative family history, atypical

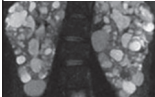
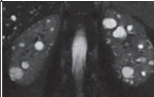
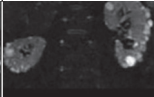
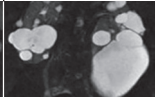
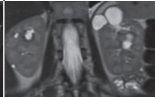
	Major genes		Minor genes					
	PKD1	PKD2	ALG5	ALG8	ALG9	GANAB	IFT140	DNAJB11
Contribution to all genetic diagnoses	~70–80%	~15–20%	<0.5%	<0.5%	<0.5%	<0.5%	~1–2%	<0.5%
Imaging pattern	Typical ADPKD		Atypical ADPKD					
Imaging case examples of cDNA and protein level mutations (eGFR=mL/min per 1.73 m ²)	Male age 37 years eGFR: 70 2113C→T, Gln705Ter	Male age 39 years eGFR: 95 1973delC, Pro658fs	Female age 57 years eGFR: 59 919C→T, Arg307Ter	Female age 52 years eGFR: 96 446T→G, Leu149Arg	Male age 29 years eGFR: 95 302A→G, Tyr101Cys	Male age 59 years eGFR: 91 1321delC, His441fs	Female age 58 years eGFR: 93 1432+1G→A	Female age 38 years eGFR: 110 267T→A, Tyr89Ter
								
Kidney phenotype	Mild to severe bilateral kidney enlargement with numerous to innumerable cysts in a symmetrical distribution; on average, most severe disease associated with PKD1 PT mutations, followed by PKD1 IF indels, PKD1 NT, and PKD2 mutations		Normal to mild kidney enlargement with a few to numerous cysts and occasionally lopsided distribution pattern				Normal to mild kidney enlargement with a few large, predominantly exophytic cysts	Normal to small kidneys with few to numerous cysts, similar to ADTKD
Kidney prognosis	Mean age of ESKD onset by mutation: 53 years (PKD1 PT), 59 years (PKD1 IF indel), 71 years (PKD1 NT), and 80 years (PKD2 mutations)		Some develop ESKD in older age (62–91 years)	Preserved kidney function in older patients	Significant CKD in older patients	Rarely progress to ESKD	Preserved kidney function in older age	Some patients develop ESKD from age 59–89 years
Liver cysts	Most patients have few to numerous liver cysts by age 40 years		Uncommon	Common			Uncommon	
Symptomatic PLD	>5% of patients, mostly female and independent of genotype		Rare, likely due to low prevalence of the cystic disease	Possible in a small percentage of patients	Rare, likely due to low prevalence of the cystic disease		Not well defined, but unlikely	
Risk of ICA	Increased		Unclear					Possibly increased

Figure 1: Clinical features of major and minor cystic genes for adult patients with ADPKD

ADPKD=autosomal dominant polycystic kidney disease. ADTKD=autosomal dominant tubulointerstitial kidney disease. CKD=chronic kidney disease. eGFR=estimated glomerular filtration rate. ESKD=end-stage kidney disease. ICA=intracranial aneurysm. IF=in-frame. NT=non-truncating. PLD=polycystic liver disease. PT=protein-truncating.

The KDIGO guideline¹¹ proposes a nomenclature for reporting mutations in clinically diagnosed ADPKD or ADPLD. For genetically screened patients, ADPKD will be used as the disease name followed by a major (*PKD1* or *PKD2*) or minor ADPKD gene locus. Similarly, ADPLD will be employed as the disease name followed by a major (*PRKCSH* or *SEC63*) or minor ADPLD gene locus.¹¹ For genetically unscreened or screened negative patients, ADPKD or ADPLD without any gene name will be used for reporting, depending on their clinical phenotype. A list of the major and minor ADPKD genes and their clinical correlates are shown in figure 1. This proposal is expected to facilitate more accurate diagnosis and improve genotype–phenotype correlations. However, it also has the potential for harm (eg, discrimination and refusal of life insurance coverage in some countries) among patients with minor gene mutations. Thus, patients and their families, health-care providers, insurance companies, and others dealing with the welfare of individuals with ADPKD need to be educated that unlike ADPKD associated with the major genes, patients with ADPKD associated with minor cystic disease genes generally have a benign clinical course and low risk for kidney failure.¹¹

ADPKD is primarily caused by mutations of *PKD1* and *PKD2*, resulting in dysfunction of the PC1–PC2 complex, triggering aberrant regulation of cellular signalling pathways that lead to cyst formation and enlargement.⁴⁶ During disease progression, increased cystic volume exerts pressure on surrounding normal kidney tissue, leading to tissue damage from tubular obstruction, decreased blood flow, and eventually kidney failure.⁴⁷ A notable feature of ADPKD is focal cyst development along any nephron segment with increasing age.^{47,48} Genetic studies of human cystic epithelia and orthologous knock-out mice support a two-hit model of cystogenesis, in which cyst formation is triggered by the complete loss of PC1 or PC2 within a tubular epithelial cell due to a germline mutation of *PKD1* or *PKD2* and a somatic mutation of *PKD1* or *PKD2*.^{49–54} However, more recent studies including patients with digenic *PKD1* or *PKD1* and *PKD2* mutations^{20,25,37–41} and *Pkd1* knock-out mice with an incomplete mRNA splicing defect⁵⁵ suggest that a threshold model of cystogenesis might better explain the cyst initiation process (ie, lowering of the normal cellular PC1 or PC2 level to ~20% could trigger cyst formation).^{46,56,57}

clinical trials that led to the approval of tolvaptan for clinical treatment in ADPKD.⁵⁸⁻⁶¹ Multiple other dysregulated signalling pathways including AMP-activated protein kinase (AMPK), mammalian target of rapamycin (mTOR), extracellular signal-regulated kinase (ERK), insulin-like growth factor -1 (IGF-1), and WNT- β -catenin have been shown to modulate cyst growth through biological processes such as cystic fluid secretion, cellular proliferation, and inflammatory responses in preclinical studies (figure 2).^{46,62} Two current areas of focus for therapeutic development are to target the cilia-dependent cyst activation pathway (appendix p 2)⁶³⁻⁶⁶ and metabolic reprogramming (appendix p 2).⁶⁷⁻⁷⁴ Additionally, in a study of mouse models of established ADPKD, regression of cystic disease was initiated when *Pkd1* or *Pkd2* are re-expressed.⁷⁵ Thus, future precision medicine

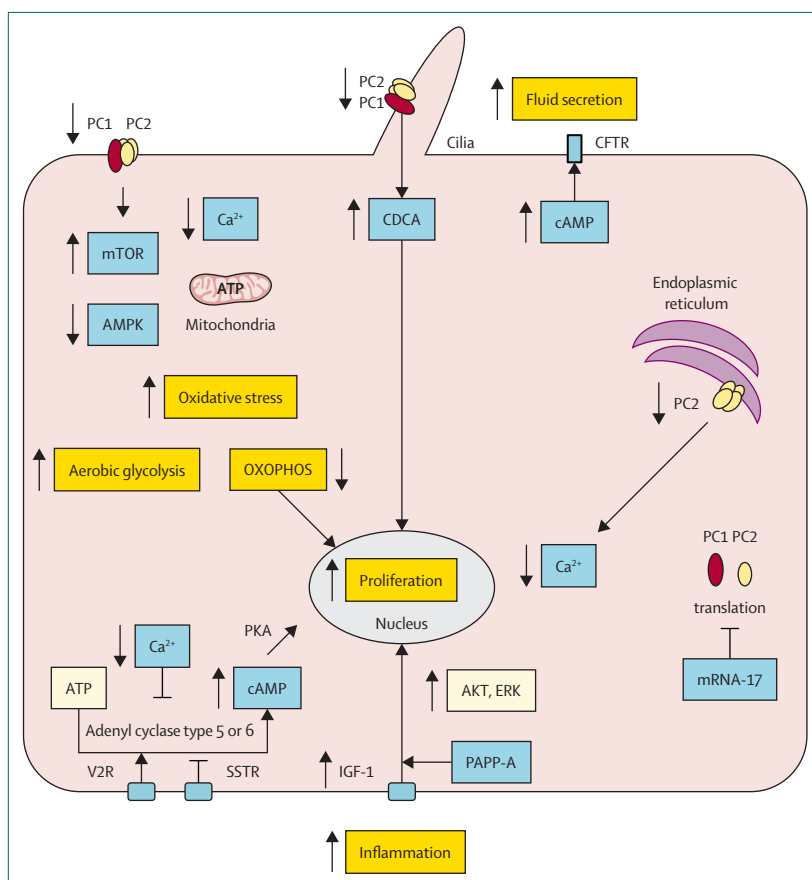


Figure 2: Major signalling pathways implicated in the pathogenesis of ADPKD

PC1 and PC2 form a mechanosensory complex located at the primary cilium and other kidney and liver tubular epithelial cellular locations, where they modulate calcium influx and integrate extracellular mechanical and chemical signals to regulate intracellular signalling cascades. Reduced intracellular PC1 or PC2 dosage below a critical threshold leads to decreased Ca^{2+} , increased cAMP, activation of mTOR and suppression of AMPK, and other signalling pathways that promote cell proliferation, fluid secretion, and inflammation. This knowledge led to the successful development of tolvaptan (V2R antagonist) and somatostatin analogues (SSTR agonists) for the treatment of cystic kidney and liver disease, respectively. Current and emerging therapeutic developments of interest include targeting metabolic reprogramming (by AMPK activation), PAPP-A (a secreted metalloproteinase that cleaves IGF binding proteins to increase local IGF-1 availability), microRNA-17 (which impairs PC1 and PC2 translation and mitochondrial metabolism), and a CDCA pathway. ADPKD=autosomal dominant polycystic kidney disease. CDCA=cilia-dependent cyst activation. OXOPHOS=oxidative phosphorylation.

approaches that correct the disease allele or protein directly, such as by gene editing, gene replacement, or corrector molecules, could potentially lead to disease regression or stabilisation (figure 2). An alternative approach being explored is to increase expression of the normal or missense mutant allele (by microRNA-17

antisense oligonucleotides).^{76,77} Table 1 shows recent and ongoing clinical trials of novel therapies in ADPKD.⁷⁸

Diagnosis

The diagnosis of ADPKD relies primarily on clinical phenotype, family history, kidney imaging, and genetic

	Design	Intervention	Study outcomes	Duration (months)	Sample size (N)	NCT or UMIN number
Phase 3						
STOP-PKD	RCT	Oral SGLT2 inhibitor (dapagliflozin 10 mg) vs placebo	Safety, tolerability, and efficacy (changes in eGFR slope and TKV)	36	420†	NCT07280585
EMPA-PKD‡	RCT	Oral SGLT2 inhibitor (empagliflozin 10 mg/day) vs placebo	Safety, tolerability, and efficacy (changes in TKV and eGFR slope); stratification by tolvaptan use	18	44§	NCT06391450
IMPEDE-PKD	RCT	Oral metformin XR (AMPK activator) vs placebo at 0.5–2 g/day	Safety, tolerability, and efficacy (changes in eGFR slope and composite outcome)	24	1174†	NCT04939935
HYDRO-PROTECT	RCT	Oral hydrochlorothiazide vs placebo at 25 mg/day in patients treated with tolvaptan	Safety, tolerability, and efficacy (changes in eGFR slope and reduction of 24 h urine volume)	39	300†	NCT05373264
Phase 2						
ABBV-CLS-628	RCT	Intravenous monoclonal anti-PAPP-A antibody (inhibition of IGF-1 signaling) vs placebo every 4 weeks	Safety, tolerability, and efficacy (changes in eGFR slope and TKV)	23	240†	NCT06902558
AGAINST-PLD‡	RCT	Subcutaneous leuporelin (3.75 mg once monthly for 3 months followed by 3-monthly injections of 11.25 mg) vs standard of care for 18 months followed by intervention	Safety and efficacy (changes in liver volumes)	36	36†	NCT05478083
Empagliflozin‡	RCT	Oral SGLT2 inhibitor (to improve kidney haemodynamic and metabolism) vs placebo at 10 or 25 mg/day	Safety and tolerability; exploratory outcome (changes in TKV)	12	50§	NCT05510115
Dapagliflozin‡	RCT	Oral SGLT2 inhibitor (dapagliflozin 10 mg) vs usual care among patients receiving tolvaptan	Completed open-labelled RCT cross-over trial that suggests that an SGLT2 inhibitor may be beneficial to slow eGFR decline and TKV growth in patients on tolvaptan; limited by a lack of a wash-out period and small sample size	6	27†	UMIN000046275
GLP-1 receptor agonist‡	RCT	Oral GLP-1 receptor agonist (tirzepatide) to improve metabolism vs placebo titrated to 10 mg/day	Safety and tolerability; exploratory outcome (changes in TKV)	12	126§	NCT06582875
AGLOW‡	Single arm	PC1 corrector (VX-407, Vertex) for subset PKD1 variants	Safety and tolerability; exploratory outcome (changes in TKV)	12	24	NCT07161037
Phase 1						
RGLS8429	Single arm	Subcutaneous injection of mRNA-17 inhibitor every 2 weeks	Safety, tolerability, and pharmacokinetics; exploratory outcome with biomarkers	4	68 (completed)	NCT05521191
GSK4771261	Single arm	Intravenous or subcutaneous monoclonal anti-PAPP-A antibody	Safety, tolerability, and pharmacokinetics	6	84	NCT06734234
PYC-003	Single arm	Single intravenous dose of a PKD1 activator; antisense oligonucleotide	Safety, tolerability, and pharmacokinetics	6	24	NCT06714006
VX-407	Single arm	PC1 corrector (for subset PKD1 variants)	Safety, tolerability, and pharmacokinetics	24 days	159 healthy volunteers (completed)	NCT06345755
ADPKD=autosomal dominant polycystic kidney disease. eGFR=estimated glomerular filtration rate. RCT=randomised controlled trial. TKV=total kidney volume. *Only trials with sufficient detail on the treatment were included. †Multicentre. ‡Unlikely to be sufficiently powered to detect a meaningful treatment effect. §Single centre.						
Table 1: Recent and current clinical trials of novel therapies for ADPKD*						

testing. Most patients with ADPKD associated with the two major genes (ie, *PKD1* or *PKD2*) will typically display bilateral and symmetrical cyst distribution and enlarged cystic kidneys in imaging after the second to third decade of life. In new patients or those at risk with a positive family history of ADPKD, kidney ultrasound is the preferred method for initial screening. Among people at risk with a positive family history of ADPKD associated with a major gene, pretest probability at birth is 50% and age-specific ultrasound criteria based on number of kidney cysts have been established for both diagnosis and disease exclusion (table 2).⁷⁹ However, in the absence of a positive family history, these diagnostic criteria are not valid since the pretest probability of ADPKD (ie, population risk of ~1 per 1000) is much lower.³ Although ultrasound is a useful initial screening test, it has reduced sensitivity for detecting smaller cysts (ie, <5–7 mm) and does not provide early detection or exclusion of disease. For younger people at risk (age 16–40 years), MRI with increased sensitivity in detecting cysts larger than 2–3 mm will enable earlier diagnosis and exclusion of ADPKD associated with the major genes. For younger people at risk with a positive family history of ADPKD, the presence of more than 10 cysts in both kidneys is sufficient for diagnosis. Less than 5 cysts in both kidneys of those older than 20 years is suggested for disease exclusion.^{11,80} For patients with a positive family history and typical kidney imaging pattern, the diagnosis of ADPKD is usually sufficient by ultrasound or MRI using age-dependent criteria. In the absence of a positive family history or typical kidney imaging pattern of ADPKD, the differential diagnosis needs to be broadened to include ADPLD and ADPKD-like phenotype from the minor cystic genes,^{11,25–32} autosomal dominant tubulointerstitial kidney disease,⁸¹ recessive forms of polycystic kidney disease,⁸² syndromic forms of polycystic kidney disease (eg, tuberous sclerosis complex, von Hippel-Lindau disease, and orofacial-digital syndrome), and acquired disorders such as acquired and lithium-associated cystic kidney disease.⁸³ Age of clinical presentation, kidney imaging patterns, and distinct extrarenal features might help to distinguish these phenocopies.¹¹ Additionally, in patients who present without a family history of ADPKD and an atypical imaging pattern (Mayo Imaging Class 2A) with no *PKD1* and *PKD2* mutation detected, somatic mosaicism should be considered.^{44,45}

Genetic testing could be informative for patients suspected to have ADPKD without an apparent family history, patients with atypical or equivocal kidney imaging findings, or those with marked variability of cystic disease severity among affected relatives including very-early onset disease and syndromic forms of polycystic kidney disease.^{84,85} Additionally, genetic testing might also be useful for disease exclusion in younger people at risk who are being evaluated as a living-related kidney donor, and for selection of

Test criterion	Diagnosis confirmed if test criterion is met in a family with:			
	known PKD1 mutation	known PKD2 mutation	unknown genotype	
Diagnosis				
15–39	>3 cysts in both kidneys	100%	100%	100%
40–59	>2 cysts in both kidneys	100%	100%	100%
≥60	>4 in each kidney	100%	100%	100%
Exclusion				
15–29	No cyst	99%	84%	91%
30–39	No cyst	100%	97%	98%
40–59	<1 cyst in both kidneys	100%	100%	100%

* A minor modification of these previously published criteria is the requirement to document a typical imaging pattern (ie, bilaterally enlarged kidneys with symmetrical cyst distribution) in the patient or an older affected relative.

Table 2: Ultrasound criteria for typical ADPKD associated with PKD1 or PKD2 mutations by age (years)*

unaffected embryos in assisted reproduction to eliminate germline disease transmission.⁸⁴ Since mutations of multiple cystic genes might mimic ADPKD, a targeted NGS panel or other clinically accredited genetic or genomic test should be used for genetic testing. Standard sequencing techniques face the challenge of distinguishing the six pseudogenes that share 97% sequence similarity with *PKD1*.⁸⁶ However, with more specific exome capture probes and advanced bioinformatics, false positive or false negative results are generally rare. Clinical genetic testing should be coupled with genetic counselling. Assignment of pathogenicity in many missense variants can be uncertain and should be classified in accordance with the American College of Medical Genetics and Genomics guidelines.⁸⁷ In a patient with typical clinical presentation of ADPKD, negative or uncertain genetic test results do not exclude an inherited form of polycystic kidney disease. Figure 3 provides a diagnostic algorithm for testing patients with kidney cysts suspected to have ADPKD that integrates the use of imaging and mutation-based testing.

Risk assessment

Clinical risk factors including male sex, early onset hypertension, gross haematuria, kidney stones, overt proteinuria, and *PKD1* protein-truncating mutations have been associated with more severe ADPKD.⁸⁸ High salt intake, smoking, high blood pressure, and possibly high protein intake are other modifiable risk factors.^{89,90} More recently, high BMI (≥25 kg/m²) has also been shown to be associated with increased total kidney volume growth but inconsistently with eGFR decline or progression to stage 5 chronic kidney disease.^{91–94} In clinical practice, three approaches have been used to identify high-risk patients with rapidly progressive disease for possible disease-modifier treatment (appendix p 3).

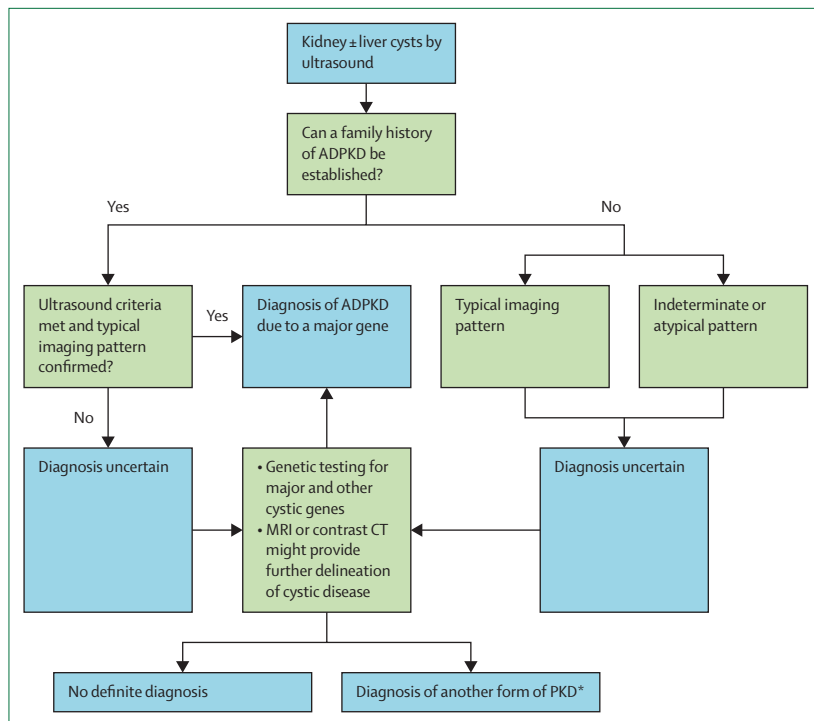


Figure 3: Diagnostic algorithm for a patient with kidney cysts suspected to have ADPKD¹¹

For diagnosis of ADPKD due to a major gene (*PKD1* or *PKD2*), confirmation for typical imaging pattern (ie, bilateral kidney enlargement with symmetrical cyst distribution) is also required in the test subject or an older affected relative. In the absence of a positive family history, the finding of a typical imaging pattern can be considered as likely diagnosis of ADPKD associated with a major gene. However, a definitive diagnosis (if preferred by the patient) will require genetic testing. ADPKD=autosomal dominant polycystic kidney disease. PKD=polycystic kidney disease. *Other non-*PKD1* and non-*PKD2* cystic disease or syndromic PKD.

Mayo Imaging Classification (MIC)

Total kidney volume as measured by MRI increases quasi-exponentially in adult life at approximately 5% per year on average and is a validated prognostic biomarker for ADPKD.^{6,95,96} Adjusted for age, MIC divides patients with a typical imaging pattern (ie, bilateral kidney enlargement with symmetrical cyst distribution) into five subclasses (1A–E), based on their annual growth rate of height-adjusted total kidney volume (<1.5%, 1.5–3.0%, 3.0–4.5%, 4.5–6.0%, and >6.0%).⁹⁷ Patients with class 1A and 1B experience slow decline in eGFR and are defined as slowly progressing when conservative measures are deemed adequate for management. Patients with class 1C, 1D, or 1E experience rapid decline in eGFR and are defined as rapidly progressing; treatment with tolvaptan should be considered. However, transition between adjacent MICs can occur in approximately 16% of patients and some patients with class 1C near the border zone with 1B might not have rapid disease progression.⁹⁷ Inspection of the kidney imaging pattern is mandatory as the prognostic prediction by MIC is not valid in patients with atypical imaging (class 2) patterns.⁹⁷ Some patients with MIC 2A (ie, unilateral, segmental, asymmetric, or lopsided polycystic kidney disease) might have somatic mosaicism

or mutations in the minor ADPKD genes, which all generally carry a favourable prognosis.^{44,45} Additionally, MIC should not be used in patients with mutations in genes other than *PKD1* or *PKD2*.¹¹

Although manual segmentation is the gold standard for total kidney volume measurement, it requires intensive use of a radiologist's time. Thus, an ellipsoid method, which requires only measurements of three orthogonal kidney axes, is commonly used in clinical practice.⁹⁸ Several research teams have recently developed AI-assisted algorithms for total kidney volume measurements by semi-automated or fully automated segmentations,^{99–102} for differentiation of MIC class 1 and 2 patterns,¹⁰³ and for second-generation kidney radiomics to improve risk prediction beyond total kidney volume.^{103–106} However, these approaches need to be validated before they can be used clinically.

When MRI or CT is unavailable or contraindicated, total kidney volume can be measured by ultrasound using the ellipsoid method. Compared with MRI, ultrasound systematically underestimates total kidney volume by 11%; thus, it provides high accuracy (92%) when classifying MIC 1C–E, but not 1A and 1B.⁹⁸ Additionally, an average kidney length of more than 16.5 cm in patients 45 years and younger has been proposed as a cut-off for identifying high-risk patients.¹⁰⁴ These methods are less accurate than MIC; they also require exclusion of the atypical imaging patterns, which is more difficult on ultrasound than MRI.¹⁰⁷

Predicting Renal Outcomes in Polycystic Kidney Disease (PROPKD) score

The presence of a modifier effect in ADPKD makes the use of mutation-based prognostication unreliable at the level of the individual patient.^{35–42} By combining mutation class (ie, *PKD1* protein truncating vs *PKD1* non-truncating vs *PKD2* mutations) with clinical risk factors including sex and the presence or absence of hypertension or urological events (ie, gross haematuria, flank pain from cystic burden, or cyst infection) before age 35 years, the PROPKD score could improve risk stratification relative to mutation class alone.¹⁰⁸ A PROPKD score of 3 or less indicates low risk and more than 6 indicates high risk for end-stage kidney disease before age 60 years. This approach requires accurate patient recall of the timing of previous clinical events and might not be helpful for patients younger than 35 years.

European Renal Association (ERA) risk assessment algorithm

Annual eGFR decline in patients with ADPKD can range from 0.63 to 4.65 mL/min per 1.73 m². The ERA risk assessment algorithm advocates the use of an upper age limit (≤55 years) and rate of eGFR decline (>3.0 mL/min per 1.73 m² over a period of at least 4 years) for identifying high-risk patients for disease-modifier treatment.¹⁰⁹ This approach is simple and inexpensive, but requires the

availability of serial serum creatinine measurements for at least 4 years and standardised creatinine measurements if different laboratories are used. Rapid eGFR decline in some patients might be confounded by the presence of a second kidney disease (eg, renovascular disease or metabolic syndrome) that might not be easily recognised in the absence of kidney imaging (ie, to identify mild cystic disease as a discordant phenotype).¹¹⁰ When the eGFR slope is unavailable or unreliable such as in stage 1 chronic kidney disease the ERA algorithm recommends using the MIC or PROPKD for risk assessment.

Management of ADPKD

General measures

All patients should be educated on conservative measures including hydration, diet, blood pressure, and weight control (figure 4). High water intake has a theoretical benefit of suppressing arginine vasopressin release to reduce kidney cyst growth, but the evidence supporting this practice is inconclusive. An RCT of 184 adults with ADPKD showed that water intake of approximately 3 L/day to maintain a urine osmolality of 270 mOsmol/kg versus ad lib intake of approximately 2.2 L/day did not slow total kidney volume growth over 3 years.¹¹¹ However, this study was limited by a small sample size and a high rate (52.3%) of patients in the prescribed water intake group unable to achieve the physiological measures of treatment adherence based on urinary osmolality. The KDIGO guideline currently recommends daily water intake of 2–3 L in patients with an eGFR of at least 30 mL/min per 1.73m² to reduce their risk of kidney stones.¹¹ Patients should also limit their salt intake to 5 g or less of sodium chloride or approximately 90 mmol of sodium per day as high salt intake is associated with more rapid eGFR decline and suboptimal blood pressure control in ADPKD.^{112,113}

The use of renin-angiotensin system (RAS) inhibitors as the first-line antihypertensive in ADPKD is supported by their kidney-protective, cardio-protective effect and overall safety profile.¹¹ The KDIGO ADPKD guideline recommends intensive blood pressure control of 110/75 mm Hg or less using RAS inhibitors for patients younger than 50 years with an eGFR of more than 60 mL/min.¹¹ This recommendation is based on the HALT-PKD RCT of 558 patients (aged 18–49 years with stage 1–2 chronic kidney disease), which showed that a lower blood pressure target (95/60–110/75 mm Hg) using home blood pressure monitoring was associated with a smaller increase in total kidney volume (5.6% vs 6.6%, $p=0.006$) and a decrease in urinary albumin (3.8% vs 2.4%, $p<0.001$), but more frequent light-headedness (81% vs 69%, $p=0.02$) compared with a higher blood pressure target (120/70 to 130/80 mm Hg).¹¹⁴

Post hoc analysis of HALT-PKD trials showed that only patients with severe disease (MIC 1D or 1E) who were randomly assigned to rigorous blood pressure control had slower total kidney volume increase, and slower decline in eGFR after 4 months of treatment compared



Figure 4: Conservative management for all patients with ADPKD

Advice and guidance regarding diet, exercise, lifestyle, and BP control are essential components of holistic care for patients with ADPKD. The KDIGO guideline¹¹ emphasises the importance of maintaining a balanced diet rich in fruits, vegetables, and whole grains; moderating total protein intake, limiting processed foods and added sugars; and preserving a healthy body weight (BMI ranging from 18.5 to 24.5 kg/m²). Incorporating structured nutritional counselling is increasingly recognised as an effective strategy to support adherence and facilitate individualised dietary planning. In addition, all people with ADPKD should be counselled to avoid all tobacco products, reduce alcohol intake to 1 or less alcoholic drink per day if female and 2 or less drinks per day if male, and avoiding excessive caffeine consumption (< 400 mg/day), as well as recreational drugs and anabolic steroids. ACEI=angiotensin-converting enzyme inhibitor. ARB=angiotensin receptor blocker. BP=blood pressure. CKD=chronic kidney disease. G=stage. SBP=systolic blood pressure.

with the standard blood pressure control group, raising the question of whether other patients (MIC 1A–C) should also have a strict blood pressure target.¹¹⁵ For patients 50 years or older or with an eGFR less than 60 mL/min, a systolic blood pressure target of less than 120 mm Hg by standardised office blood pressure measurements is recommended based on the 2021 KDIGO chronic kidney disease guideline.¹¹⁶ Dual RAS inhibitor and angiotensin receptor blockade should be avoided due to no additional benefit compared with monotherapy and potentially increased side-effects.^{114,117} While striving for optimal control, all blood pressure treatment should avoid symptoms of orthostatic hypotension.¹¹

Overweight and obesity should be avoided in ADPKD given their association with higher risk of progression. Assessment of these conditions using BMI needs to be adjusted for the cystic kidney and liver weights, which can be approximated from MRI.¹¹ Although multiple

interventions such as a ketogenic diet, metformin, sodium-glucose cotransporter-2 inhibitors, and GLP-1 receptor agonists have been used to treat diabetes, obesity, and metabolic syndrome, their efficacy for slowing progression in ADPKD is currently unknown. Therefore, these interventions should not be used for this purpose until results from adequately powered and long-term clinical trials become available (table 1).^{11,78} Finally, systematic screening for extrarenal manifestations such as liver cysts, intracranial aneurysms, and cardiac valve abnormalities and routine psychological assessment are also key for the comprehensive management of ADPKD.¹¹

Kidney replacement therapy

Kidney replacement therapy is the primary treatment for stage 5 chronic kidney disease and includes haemodialysis, peritoneal dialysis, and kidney transplantation.¹¹⁸ Despite concerns about constrained abdominal space and increased pressure due to enlarged kidneys in ADPKD, the mortality rate is similar between both dialysis techniques, with overall low certainty of evidence. Patients with ADPKD using peritoneal dialysis generally face fewer complications.^{119,120} An individualised approach with careful attention to potential risks such as abdominal hernia and leakage is recommended. Kidney transplantation is generally preferred for quality of life, especially since patients with ADPKD have better mean long-term post-transplant survival rates compared with other causes of kidney failure.¹¹ After transplantation, native kidneys can diminish in size, reducing so-called mass effect (eg, bloating, early satiety, acid reflux, abdominal fullness, and pain) symptoms.¹²¹ Native nephrectomy in patients with ADPKD receiving kidney transplantation should only be performed after shared decision making with multidisciplinary consultation, preferably during or after transplantation. Potential indications of nephrectomy include severe mass effect symptoms, recurrent kidney infection or bleeding, complicated kidney stones, intractable pain, suspected renal cell carcinoma, and insufficient space for the kidney allograft.¹¹⁸ Whenever possible, laparoscopic surgery is preferred to open surgery, and the decision between unilateral or bilateral nephrectomy should be made based on individual clinical circumstances.^{11,122}

Disease modifying treatment

Two large RCTs have shown that tolvaptan reduced kidney growth by 45% and eGFR declined by 26% in patients with early to mid-stage ADPKD over a 3-year period (TEMPO 3:4)⁶⁰ and eGFR declined by 35% in patients with later disease stages over 1 year (REPRISE).⁶¹ Collectively, these data have led to the regulatory approval of tolvaptan for treatment of ADPKD in many countries.¹¹ Tolvaptan is indicated for the treatment of patients with ADPKD who are deemed to be at high risk for progression to stage 5 chronic kidney disease generally by MIC, PROPKD score, or ERA eGFR slope algorithms (appendix

p 3). Extrapolations of the results of TEMPO 3:4 and REPRISE have given estimates that tolvaptan could delay progression to stage 5 chronic kidney disease by 7·3, 4·4, 2·9, and 1·5 years if pretreatment eGFR was 90, 60, 45, and 30 mL/min, respectively.¹¹ The KDIGO ADPKD guideline recommends the MIC as the preferred tool to identify high-risk patients for tolvaptan therapy. Non-contrast CT for total kidney volume measurement can be used for patients in whom MRI is contraindicated (eg, patients with claustrophobia or a non-removable metal part in body). Initiation of tolvaptan is a joint decision with the patient after discussion of its potential risks and benefits. Tolvaptan should be initiated with a daily dose of 45 mg upon waking and 15 mg 8 h later, titrated gradually to a maximum dose of 90 mg and 30 mg, according to individual tolerance. Regular monitoring should include kidney function, serum sodium, uric acid, and liver function tests. Patients should be educated to stop the drug during any acute illness associated with fluid loss or reduced fluid intake. Common side-effects of tolvaptan include thirst, polyuria, nocturia, and liver function test elevation.¹²³ Real life experience with tolvaptan use in patients with ADPKD indicates that polyuria up to 6–9 L per day might limit long-term treatment adherence in some patients.^{124–127} Tolvaptan-induced polyuria is influenced by urinary osmolar load; thiazide diuretics can reduce urine volume by lowering osmolality and improve tolerance.^{128,129} However, more studies are needed to evaluate the long-term effects of thiazide diuretics on GFR decline and overall kidney protection. Idiosyncratic drug-induced liver injury can occur with tolvaptan in up to 5% of patients; therefore, monthly liver monitoring during the first 18 months followed by quarterly assessments is necessary to detect liver function test changes early.^{130,131} Other side-effects of tolvaptan include hyperuricaemia and rarely, gout.¹¹

Management of other selected complications

Patients can experience various complications during their lifetime, among which kidney and liver cyst infections, pain, intracranial aneurysms, and polycystic liver disease are potentially the most clinically important.

Kidney and liver cyst infections

Cyst infection is a common complication that can lead to acute pain and sepsis in ADPKD. Patients diagnosed with suspected kidney or liver cyst infection should be promptly investigated based on history, clinical, and laboratory tests including abdominal imaging (appendix p 4).¹³² Suspected or confirmed kidney or liver cyst infection should be treated with a lipid-soluble antibiotic (ie, fluoroquinolone or trimethoprim-sulfamethoxazole) that has better cystic penetration than non-lipid soluble antibiotics for 4–6 weeks to ensure a high treatment success rate. If fever persists for more than 48–72 h, imaging tests should be repeated to rule out possible complications (such as a perirenal abscess or urinary

obstruction). In difficult cases,¹¹¹ an indium white blood cell scan or [¹⁸F]fluorodeoxyglucose-PET-CT might be useful for confirming and locating the infection. Infected cysts larger than 5 cm that are unresponsive to antibiotic treatment could require percutaneous or surgical drainage. In severe cases, surgical intervention such as nephrectomy might be necessary.¹¹ Empirical antibiotic treatment should not be prescribed for patients with acute kidney or liver pain without fever, elevated white cell counts, and CRP levels. In such cases, other causes such as cyst haemorrhage should be considered.¹¹

Pain management

Pain is a major burden in patients with ADPKD, with acute episodes often triggered by infections, kidney stones, or cyst rupture, and chronic pain originating from kidney or liver capsule distension or pressure on surrounding organs. Pain is best managed by an interdisciplinary team. Therapeutic strategies should follow a stepwise approach starting with non-pharmacologic measures, followed by oral medications (eg, non-opioid and opioids) and minimally invasive therapies (eg, foam sclerotherapy). For refractory cases, more invasive options (eg, celiac plexus block, nephrectomy, and hepatectomy) could be considered.^{133,134}

Intracranial aneurysms

The prevalence of intracranial aneurysms in patients with ADPKD is four times higher in people with a family history of intracranial aneurysm or subarachnoid haemorrhage than those without such a family history; most are small (<5 mm), predominantly located in large calibre arteries in the anterior circulation.^{135,136} Although the majority of intracranial aneurysms are asymptomatic and might not rupture,¹³⁷ rupture can result in significant morbidity and mortality. It is vital to educate patients on the increased risk and modifiable risk factors such as smoking cessation and stringent blood pressure control (appendix p 5).^{138,139} Patients with ADPKD should be educated to recognise the clinical significance of a thunderclap headache, a sudden onset, typically unilateral headache that reaches maximal intensity within seconds to minutes. This unusual headache is typically due to a minor bleed preceding intracranial aneurysm rupture and should alert the patient to seek emergency assessment immediately.¹¹ Headache in patients with ADPKD can also be due to spinal meningial diverticula or spontaneous spinal cerebrospinal fluid leak resulting in spontaneous intracranial hypotension.¹⁴⁰ Screening by time-of-flight magnetic resonance angiography or high-resolution CT angiography is recommended in high-risk individuals with a positive personal history of intracranial aneurysm or subarachnoid haemorrhage or a family history of intracranial aneurysm, subarachnoid haemorrhage, or unexplained sudden death who are eligible for treatment and have a reasonable life expectancy.^{11,138} Comprehensive discussion of the option of screening and its advantages

and limitations is important to facilitate informed decision making. Screened positive cases should be referred to the neurosurgical team for assessment and discussion of management options, including observation, coiling, clipping, and flow-diverter therapy. Rescreening every 5–10 years should be individualised for those initially screened negative but at high-risk of intracranial aneurysm.¹¹

Polycystic liver disease

Although approximately 80% of patients with ADPKD will have liver cysts by age 40 years, only 5–10% develop symptomatic polycystic liver disease.^{141,142} Symptomatic polycystic liver disease is more common in women, generally associated with a liver volume greater than 1800 mL or double the normal size, and typically presents with mass effect symptoms but not liver failure.^{141,142} Massive liver enlargement can severely affect quality of life, body image, physical activities, sexual life, and even nutritional intake. When abdominal MRI or CT is performed for ADPKD, kidney and liver images should be included to assess both kidney and liver cystic burden.¹¹ Early screening could identify individuals at increased risk of progressive polycystic liver disease, particularly young women with multiple liver cysts in whom oestrogen exposure should be avoided. Patients with symptomatic polycystic liver disease^{143,144} should be assessed for sarcopenia and malnutrition and referred to a centre of expertise for consultation and management.¹¹ For symptomatic polycystic liver disease, treatment with a long-acting somatostatin analogue can reduce liver volume by 2–8% over 6–36 months.^{145,146} Somatostatin analogues also slow the total kidney volume growth but have no effect on kidney function decline.¹⁴⁷ Depending on the size and distribution of the liver cysts, different radiological or surgical treatments can be used, including foam sclerotherapy, cyst fenestration, segmental hepatic resection, and liver transplantation (appendix p 6).¹¹ For patients with ADPKD, liver transplantation is considered when there is a massively enlarged liver combined with at least one of the following: severe portal hypertension, hepatic decompensation, malnutrition, or severe sarcopenia.¹⁴⁸ For patients with an eGFR below 30 mL/min, a combined kidney and liver transplant might be recommended. Treatment options should also take into consideration that liver volume growth tends to plateau after menopause in females.^{149,150}

The effect of polycystic liver disease on female reproductive health is considerable. Hormonal therapies, including oral contraceptives, hormone replacement therapy during menopause, and oestrogens and gestagens for in vitro fertilisation should be used with caution in patients with clinically significant polycystic liver disease, as higher oestrogen exposure has been associated with increased cyst growth. If hormonal agents are used, oestrogen-free or low-oestrogen preparations are preferred. In women with enlarged liver considering

reproductive options, referral to an expert centre for consultation is recommended.¹⁵¹

Patient-centred care

There is increasing interest in developing multi-disciplinary care pathways,¹⁵² identifying future research priorities,¹⁵³ and defining clinical trial outcomes¹⁵⁴ for ADPKD that are truly patient-centred through patient-initiated or patient-embedded studies. Clear advantages include addressing major patient priorities, patient perceptions, and the most relevant research questions, thus promoting patient engagement in self-management and research.^{155,156} The top research priority identified by a patient-initiated research priority setting partnership was to develop treatments to slow or prevent kidney function decline.¹⁵³ A second common issue was the importance of kidney pain as a neglected symptom that has a major effect on quality of life.¹⁵⁷ The optimal organisation of medical care was identified by a systematic review and ranked as the third highest priority.¹⁵³ Appendix (p 7) summarises a patient-centred care pathway model for comprehensive management of ADPKD.

Current challenges and future directions

Advances in kidney imaging and NGS-based genetic technologies over the past decade have provided clinicians with better diagnostic and prognostication tools to improve clinical management. Recognition of milder ADPKD-like phenotypes associated with several minor cystic genes requires minor modification of the established imaging-based diagnostic criteria. Although tolvaptan is currently the only approved disease-modifying treatment for patients with rapidly progressive disease, new concepts of disease mechanisms (figure 2) are leading testing of promising new therapies. Looking ahead, a patient-centred model of care together with implementation and refinement of these new clinical tools and continued development of disease mechanism-based therapeutics have the exciting potential to delay or arrest kidney failure, reduce symptom burden, and improve quality of life for patients with ADPKD.

Contributors

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